

Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

## 1. Work Integrated Learning Report

Bachelor of Science in Information Technology  
Department: Software Development

Reg. No. : 2000/000752/07

***Faculty of Information Technology***

***BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY***

***WORK INTEGRATED  
LEARNING LOGBOOK***

Registered with the Department of Education as a Private Higher Education Institution under the Higher Education Act, 1997. Registration Certification No. 2000/HE07/008



richfield.ac.za

## 2. Table of Contents

|                                                                          |           |
|--------------------------------------------------------------------------|-----------|
| <b>1. Work Integrated Learning Report .....</b>                          | <b>1</b>  |
| <b>2. Table of Contents.....</b>                                         | <b>2</b>  |
| <b>3. Introduction.....</b>                                              | <b>3</b>  |
| <b>4. Body Of Report .....</b>                                           | <b>4</b>  |
| <b>4.1 Brief History of the Organization .....</b>                       | <b>4</b>  |
| <b>4.2 Department Summary and Functions.....</b>                         | <b>5</b>  |
| <b>4.3 Tasks Achieved at the Organization .....</b>                      | <b>6</b>  |
| 4.3.1 Project Familiarization and Environment Setup .....                | 6         |
| 4.3.2 Understanding System Architecture.....                             | 6         |
| 4.3.3 Learning Development Processes.....                                | 6         |
| 4.3.4 Software Development and Bug Fixing.....                           | 7         |
| <b>4.3.5 Feature Enhancement and System Improvements.....</b>            | <b>7</b>  |
| <b>4.3.6 Testing, Deployment and Release Activities.....</b>             | <b>7</b>  |
| 4.3.7 Documentation and Reporting .....                                  | 8         |
| 4.3.8 Github .....                                                       | 8         |
| <b>4.4 Application of Theoretical Knowledge Gained at Richfield.....</b> | <b>9</b>  |
| Programming modules.....                                                 | 9         |
| Database modules.....                                                    | 9         |
| Software Engineering modules.....                                        | 9         |
| Web Development modules .....                                            | 9         |
| Systems Analysis and Design.....                                         | 9         |
| Networking and Information Security.....                                 | 10        |
| IT Project module.....                                                   | 10        |
| <b>5. Conclusion.....</b>                                                | <b>11</b> |
| <b>6. Literature Reference .....</b>                                     | <b>12</b> |
| <b>7. Addendum.....</b>                                                  | <b>13</b> |
| <b>Appendix A: Jira.....</b>                                             | <b>13</b> |
| <b>Appendix B: Docker Desktop .....</b>                                  | <b>14</b> |
| <b>Appendix C: Grafana .....</b>                                         | <b>15</b> |
| <b>Appendix D: Lens .....</b>                                            | <b>16</b> |
| <b>Appendix E: Postman.....</b>                                          | <b>17</b> |
| <b>Appendix F: DBeaver .....</b>                                         | <b>18</b> |
| <b>Appendix G: Confluence .....</b>                                      | <b>19</b> |

### 3. Introduction

Work Integrated Learning (WIL) gave me the opportunity to apply the knowledge that I gained during my studies in a real working environment. My placement was completed at Halo Dot (<https://www.halodot.io/>), a division within Synthesis Software Technologies (<https://www.synthesis.co.za/>), where I was placed in the Software Development Department.

My role during the placement was that of a Software Developer in training. I was involved in setting up development environments, analysing system architecture, exploring database structures, testing APIs, fixing software defects and implementing small system enhancements. I also participated in daily stand-up meetings, sprint planning sessions, code reviews and deployment activities.

The project environment consisted of multiple technologies including TypeScript, React, PostgreSQL, Docker, AWS cloud services, Jira and Confluence. Through exposure to these technologies I gained practical insight into how modern enterprise software systems are developed, managed and maintained.

The experience helped me understand how the concepts learned at Richfield are applied in industry and allowed me to gain practical experience using technologies and tools that are commonly used in software development organizations

## SoftPOS Solutions



### Tap on Device

Accept contactless payments in person using our SoftPOS solution

[Download Brochure](#)



### Tap on Own Device

Transforming e-commerce payments, enabling consumer to tap on their own devices to complete a transaction

[Download Brochure](#)

## 4. Body Of Report

### 4.1 Brief History of the Organization

Synthesis Software Technologies is a South African software engineering and technology consulting company that develops digital solutions for businesses in various industries. The company focuses on software development, cloud technologies, architecture, digital transformation and financial technology solutions.

Halo Dot operates within this environment and is involved in the development and maintenance of payment-related software solutions, including a SoftPOS platform. SoftPOS technology allows merchants to accept payments using mobile devices without requiring traditional point-of-sale hardware. This company reminds me of my previous modules experiences that had to do with PCI and compliances like iso27001 and PCI-PIN and PCI-DSS

The organization follows modern software engineering practices and places a strong focus on collaboration, quality software delivery and continuous improvement. The other thing that caught my eye was how intelligent the team is and how they work well together and knowledge gathering never ends. I enjoyed also that people do a lot of documentations and writing specs and tests is a top priority.



[About us](#) [Solutions](#) [Resources](#) [FAQ's](#) [Developers](#)

[Demo](#) [Contact Us](#)

## Our Offices



### Johannesburg

South Africa

Synthesis, 1st Floor,  
34 Melrose Boulevard, Melrose Arch,  
Johannesburg, 2090



### Amsterdam

Netherlands

Halo Dot BV, Fred.  
Roeskestraat 115, 1076 EE,  
Amsterdam, Netherlands



### Cape Town

South Africa

Unit A204, 2nd Floor,  
Gatehouse Building Black River Park South, Fir Street,  
Observatory, Cape Town, 7925



### Florida

USA

7901 4th ST N STE 300,  
St. Petersburg,  
FL 33702 US

## 4.2 Department Summary and Functions

I was placed in the Software Development Department. The department is responsible for developing, maintaining, testing and supporting software applications used by clients and end users.

So the department has people working in the backend that process the transaction through connecting to the banks and other institutions like bankserv. There is a frontend team that deals with the app and ensuring the app is of great quality then there is another team working on the portals to ensure we have a way in which the clients can see transaction history and how to setup. Then we have a team working on the SDK the deals with the card processing. This sdk is the main part of the business as it what is costing to get certifications. There is also a branding team that works on ensuring we have quality assets like images and colors that we use to look great. There is the team of product owners and sales that work together to bring in the business.

The department for software development consists of:

- team leads, these are leads for each of the departments, they are sponsible for team growth and planning and unblocking and ensuring that there is autonomy in the team. They are checkers of timesheets and quality of work.
- Software developers, these are responsible for writing the code that keeps the systems running. This is the areas I was working in. we wrote code, review, document and proofread the contents.
- Testers, these groups of people are responsible for checking the quality of the code and ensuring it works as expected. They find the bugs and write test cases.
- Support Engineers, these are the first line of defense, whenever clients have issues, they reach these guys first and then they work on trying to resolve and if they fail, they send to developers.
- product owners are responsible for bridging the gap between developers and client, to ensure that there is no ambiguity in expectation and delievery.

### 4.3 Tasks Achieved at the Organization

During my placement at Halo Dot, I was exposed to various activities that allowed me to gain practical experience in software development and enterprise systems.

#### 4.3.1 Project Familiarization and Environment Setup

My first responsibility was to understand the project environment and become familiar with the technologies used by the development team. I attended project walkthrough sessions where the team explained the purpose of the SoftPOS platform, current deliverables and outstanding work.

I installed and configured development tools including Visual Studio Code and DBeaver. I cloned both the frontend and backend repositories and spent time understanding the project documentation, package configuration files and README instructions. I also learned how AWS access and permissions are managed within the organisation.

A big part of my first week involved learning Docker and understanding how multiple services work together within a development environment. I successfully built and started the project ecosystem using Docker Compose and observed how the React portal, PostgreSQL database and backend services interacted with one another.

#### 4.3.2 Understanding System Architecture

After setting up the environment, I spent time understanding the architecture of the application. I identified the main components of the system, including the authentication service, transaction service and portal API.

Using DBeaver, I explored the PostgreSQL database structure and analysed the relationships between tables. I also learned how SoftPOS configurations are stored and managed within the system.

The team introduced me to the authentication process used by the application and explained how backend services communicate with frontend and mobile applications

#### 4.3.3 Learning Development Processes

During the placement I was introduced to Agile software development practices used within the organisation. I participated in daily stand-up meetings, sprint planning sessions and sprint retrospectives.

I learned how Jira is used to create, assign and track tasks, and how Confluence is used to store project documentation and requirements. I also gained an understanding of how

software development teams communicate with stakeholders and manage client-related issues and escalations.

#### **4.3.4 Software Development and Bug Fixing**

I was assigned my first development task involving a validation issue on a portal configuration form. To complete the task, I investigated the issue, reviewed the relevant React code and identified the cause of the problem.

I created the required Jira ticket, created a feature branch and implemented the necessary changes. After testing the solution, I submitted a pull request for review. The pull request was approved and merged into the development branch.

This task provided practical experience in debugging, version control and the code review process.

#### **4.3.5 Feature Enhancement and System Improvements**

During the third week I was assigned a feature enhancement involving the addition of a new configuration field within the SoftPOS portal.

To complete this task, I updated database migration files, modified TypeScript models and updated backend API endpoints. I then implemented the required changes on the React frontend and connected the new field to the existing application workflow.

I tested the functionality using Postman and Docker Compose to verify that the changes were working correctly across all services. During testing I encountered a Cross-Origin Resource Sharing (CORS) issue, which required further investigation and debugging before the feature could be completed successfully.

The changes were reviewed by senior developers, corrected where necessary and ultimately merged into the development branch.

#### **4.3.6 Testing, Deployment and Release Activities**

Towards the end of the placement I participated in User Acceptance Testing (UAT) and release activities. I assisted with testing new functionality, validating software changes and performing regression testing to ensure that changes did not negatively impact existing functionality.

I also observed the release process used by the organisation. This included the creation of release branches, deployment to production environments and post-deployment verification testing.

Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

This experience provided valuable insight into software release management and the importance of quality assurance in enterprise software development.

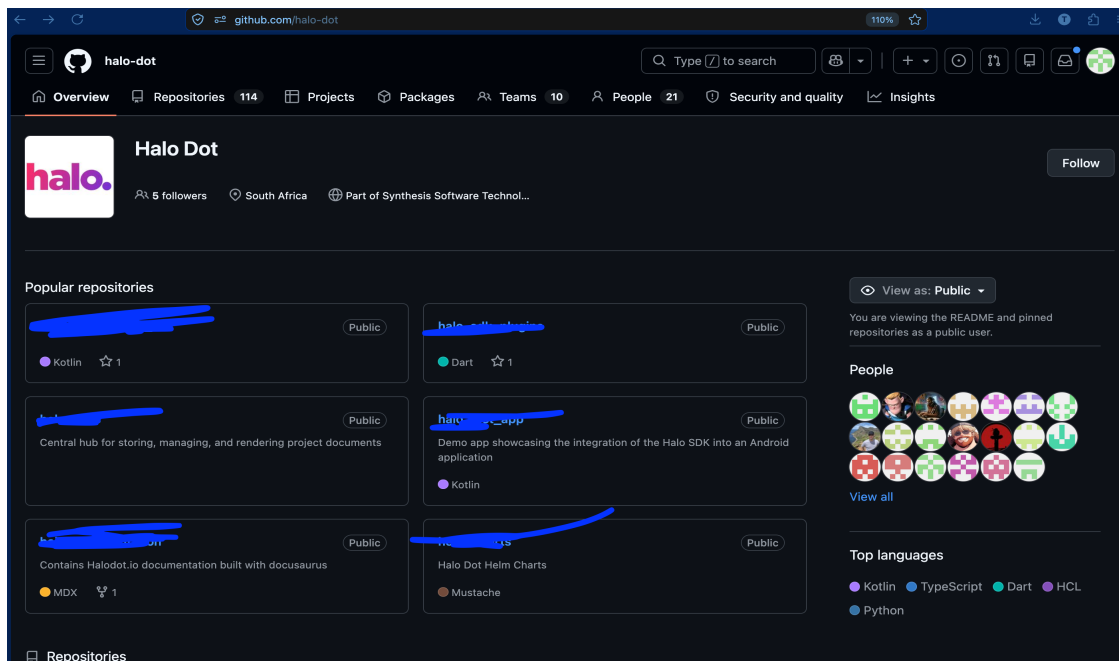
#### 4.3.7 Documentation and Reporting

Throughout the placement I documented work completed through Jira tickets and comments. I also prepared technical notes explaining changes made during development tasks and maintained records of activities completed during the placement period.

This helped me understand the importance of documentation, accountability and traceability within software development projects.

#### 4.3.8 Github

This too became my best friend as I used it to store code and also its used document in .md files, we also used it for testing through pipelines and automation was achieved with github. Through github we could also review code and writes comments and suggestions.





#### 4.4 Application of Theoretical Knowledge Gained at Richfield

One of the main objectives of the Work Integrated Learning programme is to allow students to apply theoretical knowledge gained during their studies in a real working environment. During my placement at Halo Dot, I was able to apply several concepts learned throughout my Bachelor of Science in Information Technology qualification.

**Programming modules** completed at Richfield helped me understand the TypeScript codebase used by the organisation. Concepts such as variables, functions, classes, object-oriented programming and software logic enabled me to follow the flow of the application and understand how different components interact with one another. This knowledge assisted me when investigating bugs, reading source code and implementing system changes.

**Database modules** provided a solid understanding of relational database concepts, including tables, primary keys, foreign keys and relationships between data entities. This knowledge was useful when exploring the PostgreSQL database using DBeaver, analysing data structures and making database-related changes as part of a feature enhancement. Understanding database design principles helped me better understand how information is stored and managed within the application.

**Software Engineering modules** was applied throughout the placement. Concepts such as the Software Development Life Cycle (SDLC), requirements gathering, testing, debugging, documentation and version control were all visible in the organisation's development processes. I was able to see how software moves through different stages, from planning and development to testing and deployment.

**Web Development modules** helped me understand how the React frontend communicates with backend APIs. Concepts such as client-server architecture, HTTP requests, APIs, form validation and data handling were applied while working on portal-related tasks and testing API endpoints using Postman.

**Systems Analysis and Design** concepts assisted me in understanding project requirements, acceptance criteria and business processes. Before implementing changes, I learned the importance of understanding the problem, analysing requirements and ensuring that solutions meet business needs. This was particularly useful when working on configuration enhancements and bug fixes.

Project Management concepts were also applied during the placement. The organisation follows Agile development practices and uses Jira to manage tasks and sprint activities. Through participation in daily stand-up meetings, sprint planning sessions and task management activities, I was able to see how project management principles are used to organise and deliver software projects effectively.

**Networking and Information Security** concepts gained during my studies helped me understand authentication processes, API communication and cloud-based systems. Exposure to technologies such as AWS services, JWT authentication and secure access controls demonstrated how security principles are implemented within enterprise software applications.

Overall, the workplace experience showed me how the theoretical concepts learned at Richfield are applied in a professional software development environment. It reinforced my understanding of the material covered during my studies and provided practical experience that will assist me in my future career as a software developer.

**IT Project module**, this drove me to be a very informative team for the final year project because I started implementing most of the principles I have learnt. This include git and some of the key technologies I encountered included:

- Express.js for building REST APIs
- Sequelize for database access and object-relational mapping
- PostgreSQL as the primary database
- Redis for caching and session management
- AWS services such as Key Management Service (KMS) and Secrets Manager
- Passport and JWT for authentication and authorisation
- Docker for containerisation
- Jira and Confluence for project management and documentation
- JSON Web Tokens (JWT)

I learned the purpose of common project folders such as:

| Folder                | Purpose                               |
|-----------------------|---------------------------------------|
| - <b>controllers</b>  | Handles incoming API requests         |
| - <b>services</b>     | Contains business logic               |
| - <b>repositories</b> | Database interaction layer            |
| - <b>models</b>       | Database models and entities          |
| - <b>middleware</b>   | Authentication and request processing |
| - <b>routes</b>       | API endpoint definitions              |
| - <b>config</b>       | Environment and application settings  |
| - <b>migrations</b>   | Database schema changes               |
| - <b>tests</b>        | Automated testing                     |
| - <b>utils</b>        | Shared helper functions               |

## 5. Conclusion

The Work Integrated Learning programme provided me with an opportunity to bridge the gap between academic studies and the professional working environment. Through my placement at Halo Dot (Synthesis Software Technologies), I was able to apply theoretical knowledge gained at Richfield within a real software development team and gain practical experience using industry-standard technologies, tools and processes.

The placement challenged me in several ways. Setting up the development environment, understanding a large enterprise codebase and learning unfamiliar technologies required patience, self-learning and continuous problem-solving. Overcoming these challenges helped me become more confident when working with complex systems and strengthened my ability to adapt to new technologies and working environments.

In addition to technical skills, I developed important professional skills including communication, teamwork, time management, accountability and collaboration. Working within an Agile development environment taught me the importance of documentation, code reviews, testing and effective communication between team members. One of the most valuable lessons I learned was that successful software development depends not only on technical ability but also on teamwork, planning and adherence to established processes.

The experience provided valuable insight into how enterprise software systems are designed, maintained and deployed. It also highlighted areas where I would like to continue growing, particularly in cloud technologies, software architecture and large-scale system design. At the same time, it reinforced the value of the knowledge gained during my studies and demonstrated how academic concepts are applied to solve real business problems.

Overall, this placement was one of the most rewarding experiences of my studies. It strengthened my interest in software engineering, increased my confidence as an aspiring software developer and provided practical industry exposure that will benefit me throughout my future career in the Information Technology sector. I am grateful for the opportunity to learn from experienced professionals and contribute to real-world software development projects.

Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

## 6. Literature Reference

Atlassian. 2026. Jira Software Documentation. Available at: <https://www.atlassian.com/software/jira> (Accessed June 2026).

Atlassian. 2026. Confluence Documentation. Available at: <https://www.atlassian.com/software/confluence> (Accessed June 2026).

Docker Inc. 2026. Docker Documentation. Available at: <https://docs.docker.com> (Accessed June 2026).

Grafana Labs. 2026. Grafana Documentation. Available at: <https://grafana.com/docs> (Accessed June 2026).

Lens. 2026. Lens Kubernetes IDE Documentation. Available at: <https://k8slens.dev> (Accessed June 2026).

PostgreSQL Global Development Group. 2026. PostgreSQL Documentation. Available at: <https://www.postgresql.org/docs> (Accessed June 2026).

Postman Inc. 2026. Postman Learning Centre. Available at: <https://learning.postman.com> (Accessed June 2026).

Richfield Graduate Institute of Technology. 2026. Work Integrated Learning Logbook.  
Johannesburg: Richfield Graduate Institute of Technology.

Synthesis Software Technologies. 2026. Internal Company Documentation, Project Specifications, Jira Boards, Confluence Pages and Technical Learning Resources.  
Johannesburg: Synthesis Software Technologies.

## 7. Addendum

### Appendix A: Jira

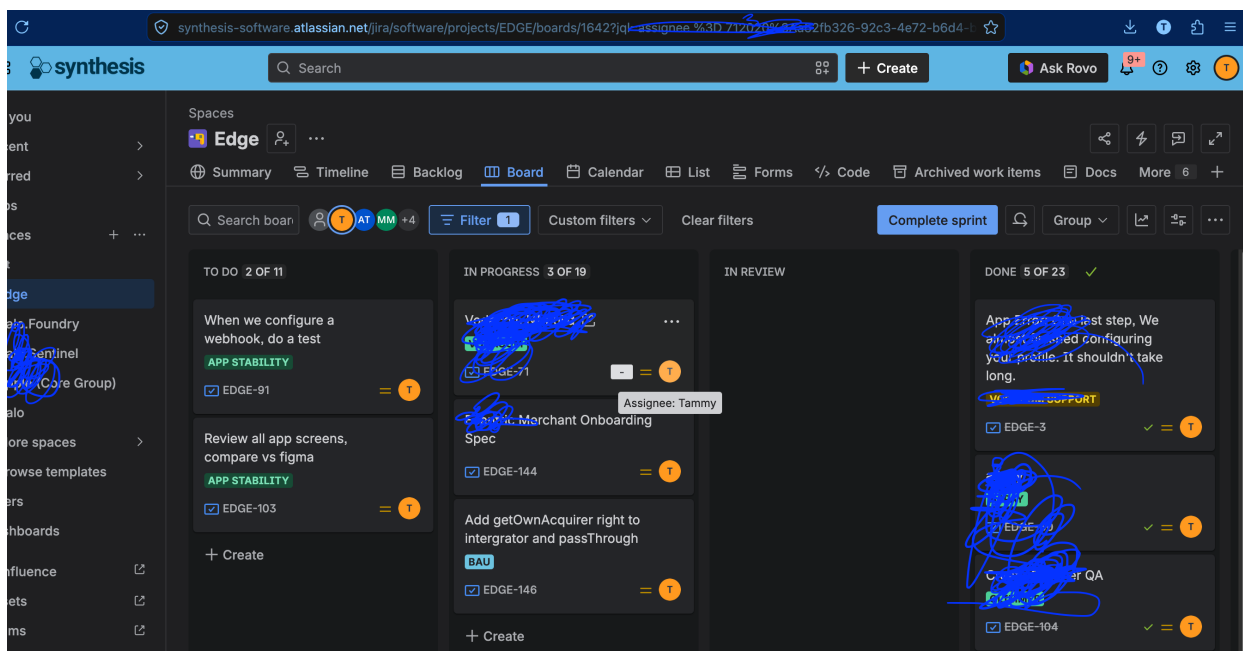
Jira is a project management and issue-tracking tool used to manage software development activities.

Throughout my placement, Jira was used to manage tasks, track progress and organise sprint activities. Every development task was linked to a Jira ticket, ensuring that work could be tracked and reviewed.

I learned how to:

- Create tickets
- Update ticket status
- Log work performed
- Participate in sprint planning
- Follow Agile development workflows

Working with Jira gave me practical exposure to Agile project management methodologies used within professional software development teams.



Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

## Appendix B: Docker Desktop

Docker is a containerization platform that allows applications and their dependencies to run consistently across different environments.

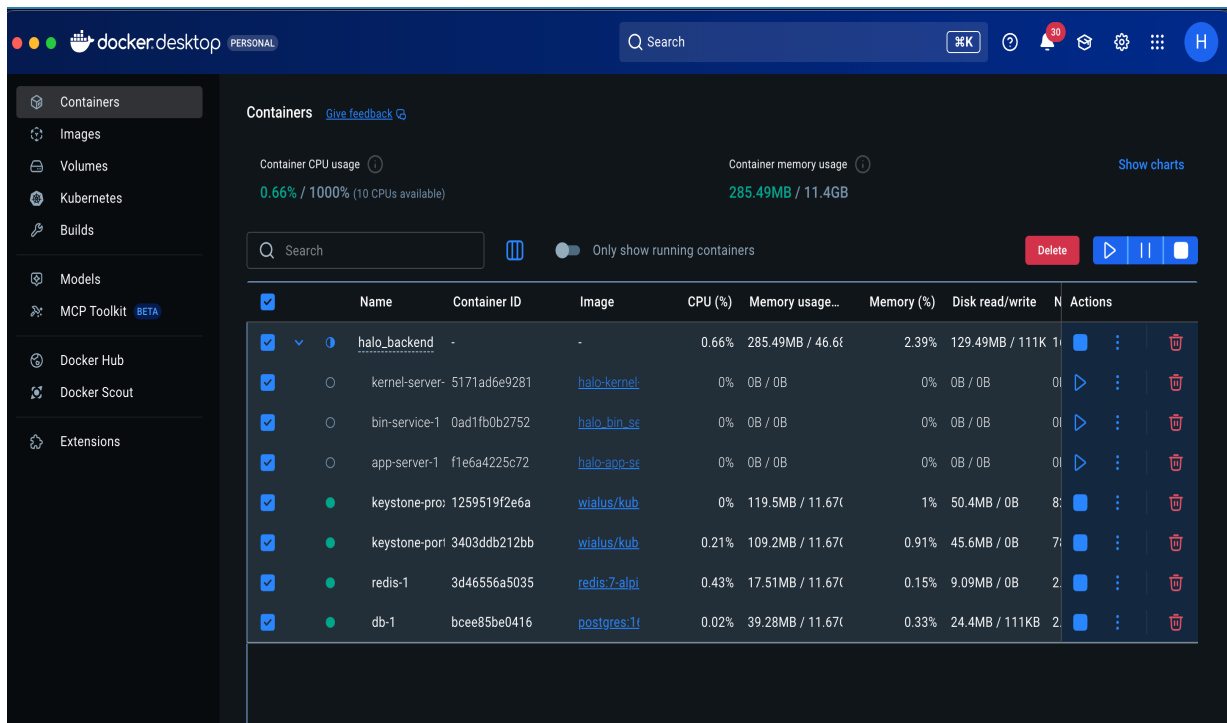
During my placement at Halo Dot, Docker was used to run the various components of the SoftPOS system locally on my machine. Instead of manually installing every service, Docker allowed the team to package applications and their dependencies into containers.

The screenshot above shows several running containers, including:

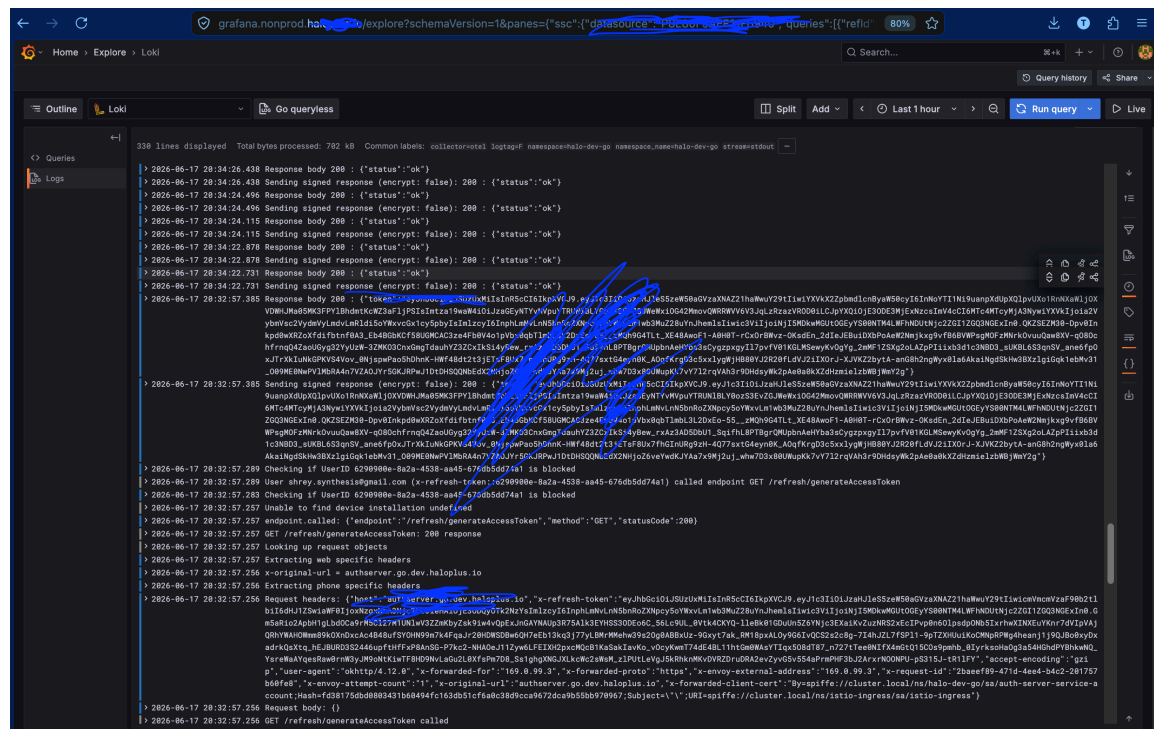
- PostgreSQL database container
- Redis container
- Authentication services
- Backend services
- FrontEnd services

I used Docker extensively during my placement to start and stop the development environment using Docker Compose commands. This allowed me to run the backend services, database and supporting infrastructure required by the application.

Through using Docker, I gained a better understanding of containerization, service orchestration and modern software deployment practices.



Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026



Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

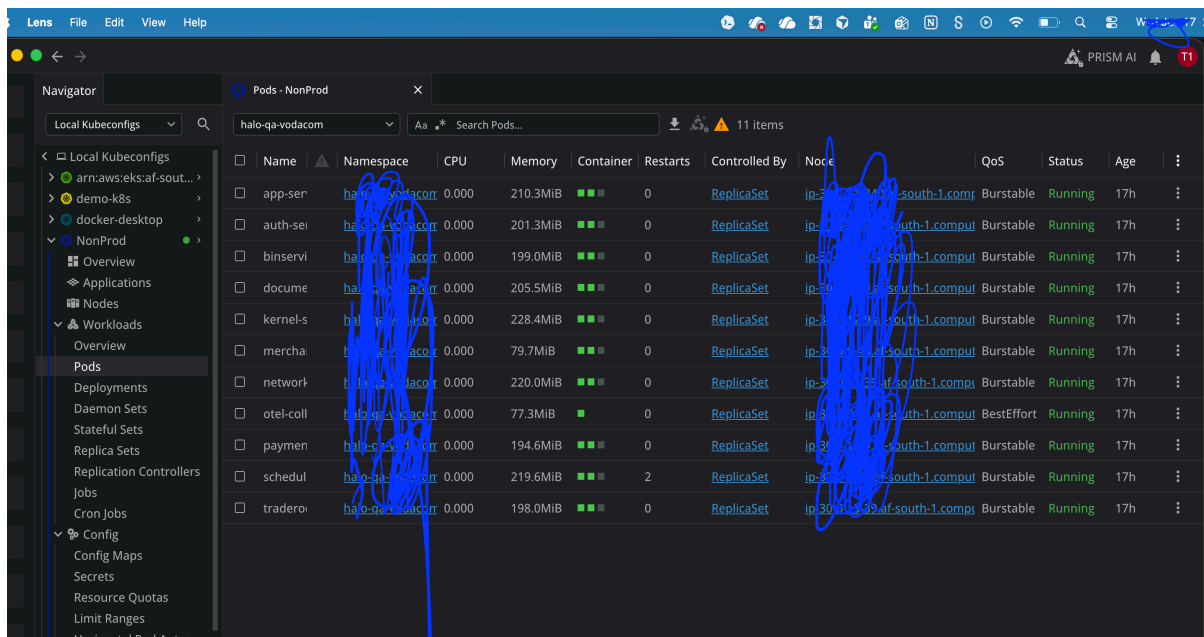
## Appendix D: Lens

Lens is a graphical user interface used to manage Kubernetes clusters.

During my placement I was introduced to Lens as a tool used to view and manage applications deployed within Kubernetes environments. Lens provides visibility into running workloads, pods, services and deployment configurations.

Although I was primarily observing and learning, the tool helped me understand how modern cloud-native applications are deployed and managed at scale.

This exposure introduced me to Kubernetes concepts and container orchestration technologies commonly used in enterprise environments.





Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

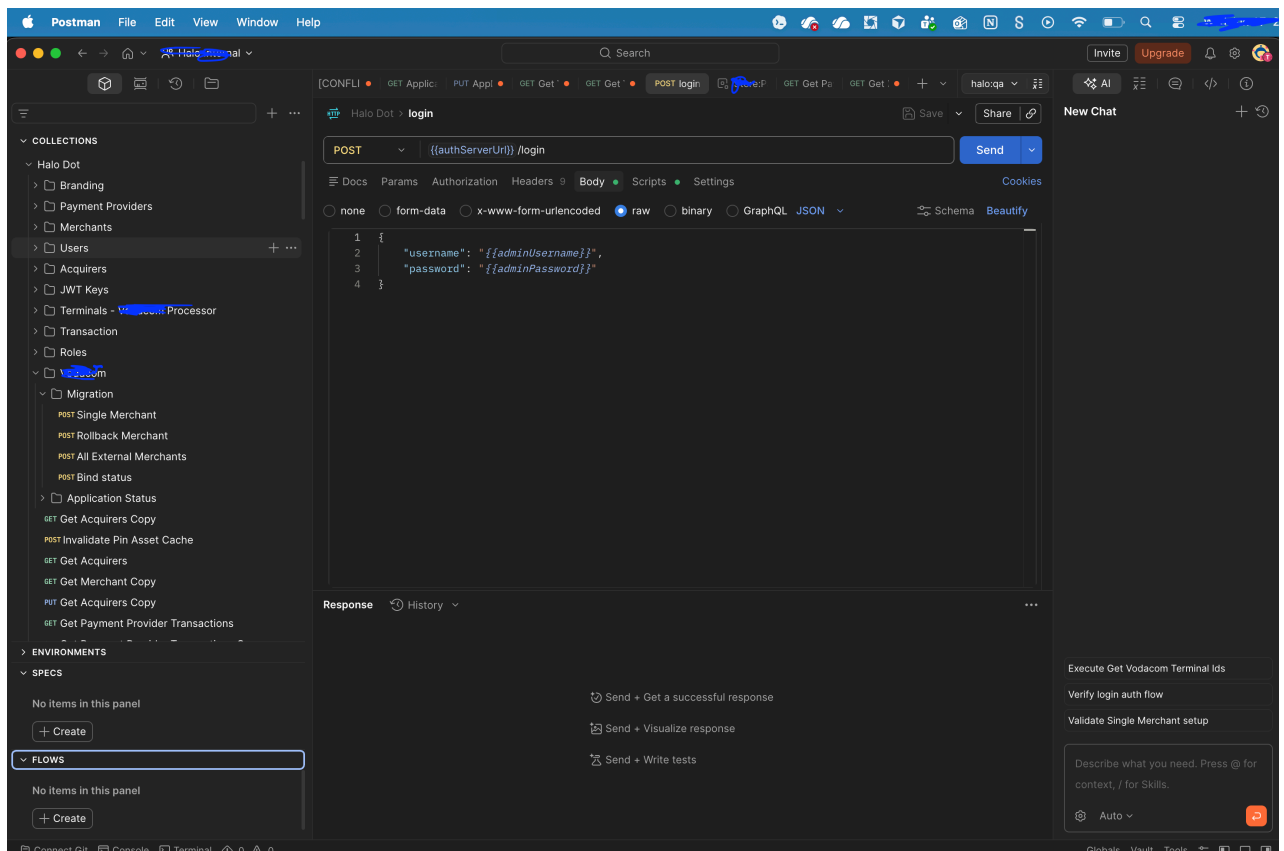
Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

## Appendix E: Postman

Postman is an API testing tool used by developers to send requests to backend services and inspect responses.

During my placement, I used Postman to test API endpoints while working on bug fixes and feature enhancements. This allowed me to verify that backend services were functioning correctly and that data was being stored and retrieved as expected.

Using Postman helped me understand REST APIs, request methods, response codes and backend debugging techniques.



Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

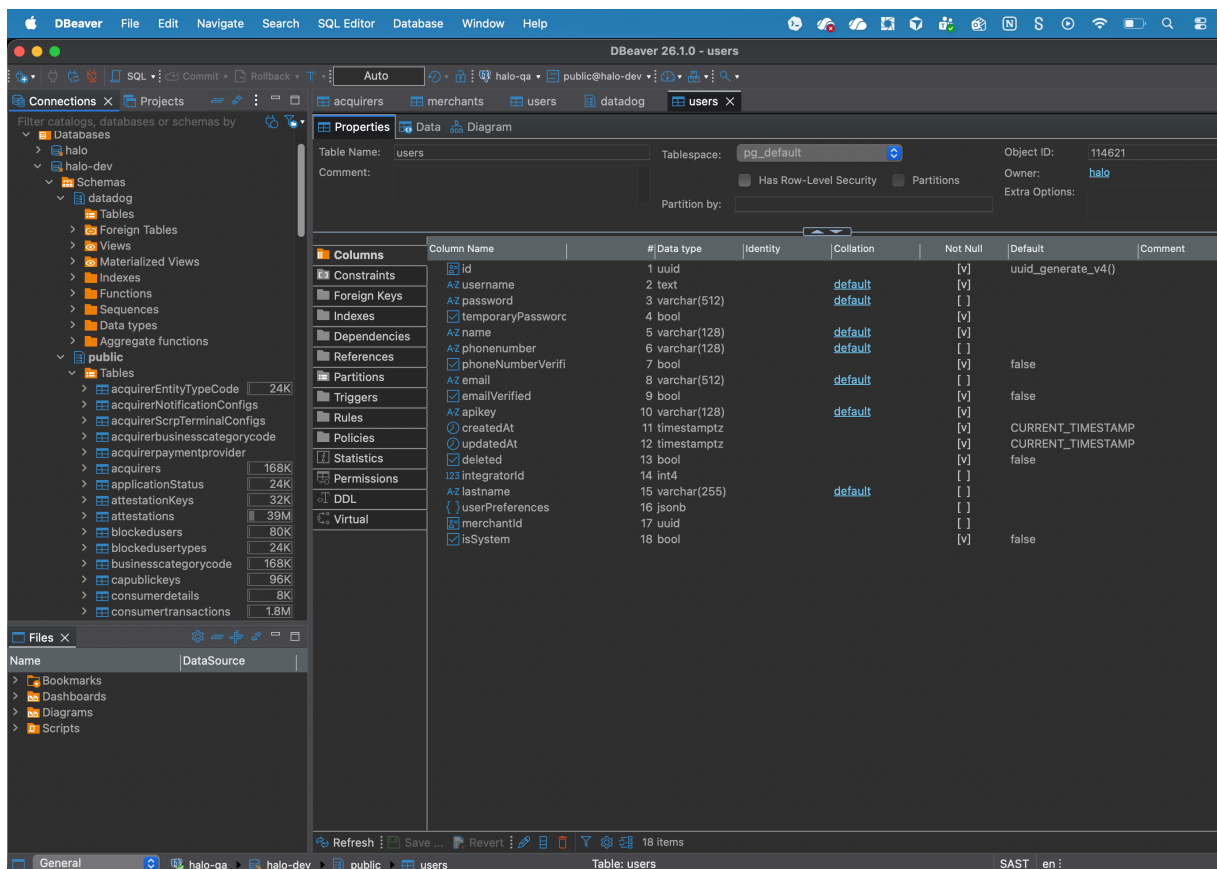
## Appendix F: DBeaver

DBeaver is a database management application used to access and manage database systems.

I used DBeaver to explore the PostgreSQL database used by the SoftPOS platform. This allowed me to view tables, relationships, records and database structures.

The tool helped me understand how application data is stored and how backend services interact with databases.

Knowledge gained from database modules at Richfield assisted me in understanding the relationships between tables and the overall database structure.



Student Name: Rivalani Tammy Nkuna  
ITS Number: 402308997

Organisation: Halo Dot (Synthesis Technologies)  
Training Period: 20 May 2026 – 12 June 2026

## Appendix G: Confluence

Confluence is a collaboration and documentation platform used to store project information, technical documentation and organisational knowledge.

During my placement, Confluence was used to access project specifications, technical documentation, requirements and onboarding information.

I learned that documentation plays a critical role in software development, especially when working within large teams and complex systems.

Confluence allowed me to understand project requirements, system functionality and development standards more effectively.

